

Leakage Prevention: Identity Recognition, Invariance Tests and Unseen Hospital Evaluation

[Author names to be added]

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Research prototype. All results are pipeline demonstrations, not clinical validation. Identity probe accuracies are means over five cross validation folds; chance is 0.050 (one in twenty training records).

1 Does the model recognize patients instead of deterioration?

We test this directly with a linear identity probe. For each cross validation fold, the trained model encodes every training window into its context vector, a logistic regression is fitted on 80 percent of those windows to classify the patient, and the held out 20 percent are scored. A model whose representation is invariant to patient identity would score at chance. Table 1 reports the probe for all eight trained variants.

Table 1: Linear identity probe on the context vector (chance 0.050).

Variant	Probe accuracy
FEAN, concatenation (baseline)	0.423
FEAN, identity adversarial (gradient reversal)	0.470
FEAN, identity adversarial + PCGrad	0.438
FEAN, gated fusion + masks	0.402
FEAN, static modality weights + masks	0.354
FEAN, modality attention + masks	0.143
Ablation: engineered features only	0.130
Ablation: embeddings only	0.439

The answer is yes, with two qualifications. The models do recognize patients: every embedding based variant scores well above chance, so leakage is present and quantified rather than assumed. But the identity signal is carried almost entirely by the embedding block: the probe falls from 0.439 to 0.130 when the ECG FM embeddings are replaced by engineered features alone. Modality attention is the only architecture that substantially reduces leakage (0.143), and it is also the best held out variant at 6 hours (AUROC 0.473). The two observations agree: identity laden embeddings hurt generalization, and mechanisms that down weight them help.

2 Identity invariance tests

Two forms of invariance testing are in place.

Measurement. The linear probe above is the invariance test, computed per fold for every variant, so invariance is a measured quantity, not an assumption.

Enforcement. Identity adversarial training (a gradient reversal layer against a patient ID classifier, optionally with PCGrad across task gradients) was added to enforce invariance. The honest result: it improves held out AUROC slightly at both horizons but does not reduce the

probe (0.423 to 0.470), so gradient reversal did not remove the leakage on this data. This negative result is reported in the manuscript rather than hidden. A natural next step, not yet run, is to combine the two mechanisms that do work: modality attention, which down weights the identity carrying embeddings, together with an identity adversary applied to the embedding block only.

3 Unseen hospital evaluation

What is covered today:

- Unseen patients: all training and evaluation splits are grouped by patient, and a patient never appears in more than one split (asserted in code).
- Unseen domain: the synthetic trained model is applied zero shot to the real Sudden Cardiac Death Holter Database and scores at chance (1 hour AUROC 0.338, 6 hour AUROC 0.447), which quantifies the synthetic to real gap.

What is not covered: an evaluation on a second hospital. The Sudden Cardiac Death Holter Database is the only open access dataset with hours of pre arrest recording, so no second open site exists for this task, and the test cannot be run on open data alone.

The planned unseen hospital evaluation is the credentialed MIMIC III Ext CA set: 36 arrest episodes in 31 ICU patients with photoplethysmogram plus electrocardiogram or arterial pressure, a different site, era, monitor type and modality set from the Holter database. The loader is written and self tested (window manifest with resuscitation exclusions, patient level splits, loud failure until credentialed access lands); the evaluation itself is pending access approval and will be reported with the same patient level, confidence interval aware protocol.

4 Summary

Patient recognition is measured, not assumed: the models recognize patients, the identity signal lives in the embedding block, gradient reversal does not remove it, and modality attention does. Unseen patient and unseen domain evaluation exist; unseen hospital evaluation is designed, loader ready, and blocked only on credentialed data access.